

GlucoSense Pro - Software Requirements Specification for Continuous Glucose Monitor

Document Control

- **Product:** GlucoSense Pro CGM Medical Device
- **Version:** 1.0
- **Date:** October 22, 2025
- **Classification:** Internal Development / FDA Class II Device
- **Author:** GlucoSense Engineering Team

System Overview

GlucoSense Pro is a Class II medical device consisting of a wearable subcutaneous glucose sensor, companion mobile app, and cloud backend for continuous glucose monitoring. The system provides real-time glucose readings, trend analysis, alerts, and data sharing for diabetes management.

Target Users

- Patients with Type 1 or Type 2 diabetes
- Healthcare providers (endocrinologists, primary care)
- Family caregivers for alert notifications

Key Features

- 14-day sensor wear with interstitial fluid sampling
- Bluetooth Low Energy (BLE) communication to mobile app
- Real-time glucose display and 1/5/15-minute trends
- Customizable high/low glucose alerts
- Cloud synchronization for long-term data storage
- Over-the-air (OTA) firmware updates
- Integration with insulin pumps and CGM ecosystems

Regulatory Context

Applicable Regulations

- **FDA 21 CFR Part 820:** Quality System Regulation for medical devices
- **IEC 62304:** Medical device software – software life cycle processes
- **ISO 14971:** Application of risk management to medical devices
- **FDA Cybersecurity Guidance for Medical Devices** (2023)
- **FDA Section 524B:** Cybersecurity requirements including SBOM submission

Compliance Standards

- ISO 13485 quality management system
- Planned 510(k) submission for predicate device equivalence

Functional Requirements

REQ-DEV-001: Sensor Hardware Integration

Priority: Critical

Description: The system shall interface with the subcutaneous glucose sensor via BLE for continuous data acquisition.

Acceptance Criteria:

1. Sensor calibration every 12 hours
2. Data transmission every 5 minutes
3. Battery life monitoring and low-battery alerts **Compliance Standards:** IEC 62304 Clause 5.2

REQ-DATA-001: Glucose Data Processing

Priority: Critical

Description: The system shall process raw sensor data into glucose readings.

Acceptance Criteria:

1. Glucose readings in mg/dL with $\pm 10\%$ accuracy
2. Trend analysis for 1/5/15-minute intervals
3. Data stored locally on device for 24 hours **Compliance Standards:** IEC 62304 Clause 5.2

REQ-ALERT-001: High/Low Glucose Alerts

Priority: Critical

Description: The system shall generate alerts for out-of-range glucose levels.

Acceptance Criteria:

1. User-configurable thresholds (e.g., < 70 mg/dL, > 180 mg/dL)
2. Audible and haptic alerts on mobile app
3. Push notifications to caregivers **Compliance Standards:** ISO 14971 Clause 7

REQ-CLOUD-001: Cloud Data Synchronization

Priority: High

Description: The system shall synchronize glucose data to a cloud backend.

Acceptance Criteria:

1. Data uploaded every 15 minutes
2. AWS-based storage with patient ID association
3. Data accessible via provider portal **Compliance Standards:** HIPAA §164.312(a)(2)(iv)

REQ-OTA-001: Over-the-Air Updates

Priority: Medium

Description: The system shall support OTA firmware updates for the sensor and app.

Acceptance Criteria:

1. Updates delivered via mobile app
2. User confirmation required before installation **Compliance Standards:** 21 CFR 820.70

REQ-QMS-001: Document Control

Priority: High

Description: Engineering documents shall be versioned and approved before implementation.

Acceptance Criteria:

1. Use Git for version control
2. Manual approval via email **Compliance Standards:** 21 CFR 820.40

REQ-CYBER-001: Software Bill of Materials (SBOM)

Priority: Medium

Description: The system shall document third-party components used in software.

Acceptance Criteria:

1. Internal list maintained in spreadsheet **Compliance Standards:** FDA Section 524B

REQ-CYBER-002: Bluetooth Communication

Priority: Critical

Description: The system shall use BLE for sensor-to-app communication.

Acceptance Criteria:

1. BLE 5.0 protocol support
2. Data transmission every 5 minutes
3. Pairing via QR code scanning **Compliance Standards:** FDA Cybersecurity Guidance

REQ-CYBER-003: Network Security

Priority: High

Description: The system shall implement network security for cloud connectivity.

Acceptance Criteria:

1. HTTPS for all API calls
2. Firewall rules for cloud servers **Compliance Standards:** FDA Cybersecurity Guidance

REQ-VALID-001: Software Verification

Priority: High

Description: The system shall undergo functional testing before release.

Acceptance Criteria:

1. Unit tests for critical components
2. Integration tests for app-sensor communication **Compliance Standards:** 21 CFR 820.30, IEC 62304 Clauses 5.6, 5.7

REQ-ANOM-001: Software Anomaly Handling

Priority: Medium

Description: The system shall log software errors for debugging.

Acceptance Criteria:

1. Error logs stored locally on app
2. Errors reported to developers via email **Compliance Standards:** IEC 62304 Clause 9

REQ-RISK-001: Risk Management

Priority: Critical

Description: The system shall identify critical risks during development.

Acceptance Criteria:

1. Risk brainstorm session conducted
2. High-level risk list maintained **Compliance Standards:** ISO 14971

REQ-CYBER-004: Penetration Testing

Priority: Medium

Description: The system shall undergo security testing before release.

Acceptance Criteria:

- 1. Annual security audit by external vendor **Compliance Standards:** FDA Cybersecurity Guidance

REQ-POST-001: Post-Market Surveillance

Priority: Medium
Description: The system shall collect user feedback post-launch.
Acceptance Criteria:

- 1. In-app feedback form
- 2. Customer support email **Compliance Standards:** 21 CFR 820.198, ISO 14971 Clause 9

REQ-CONFIG-001: Configuration Management

Priority: Medium
Description: The system shall track software versions.
Acceptance Criteria:

- 1. Version numbers assigned manually **Compliance Standards:** IEC 62304 Clause 5.1.9

Non-Functional Requirements

Performance Requirements

- Glucose reading latency: <1 second
- System availability: 99.99% for critical components
- OTA update installation: <5 minutes

Security Requirements

- HTTPS for cloud communication
- App authentication via username/password
- Secure storage of API keys

Usability Requirements

- WCAG 2.1 Level A compliance
- Intuitive app interface requiring minimal training

Data Requirements

Data Collection

- Glucose readings (timestamped, mg/dL)
- Patient demographics (linked via ID)
- Device metadata (battery, calibration status)

Data Retention

- Glucose data: 7 years per medical records laws
- Logs: 5 years
- Backup retention: 30 days

Regulatory Compliance Checklist

Requirement	Standard	Status	Notes
Document control	21 CFR 820.40	✗ Missing	No formal procedures

V&V plan	21 CFR 820.30	✗ Missing	No formal V&V plan
Change control	21 CFR 820.70	✗ Missing	No procedures defined
Usability validation	21 CFR 820.30	✗ Missing	No validation process
Complaint handling	21 CFR 820.198	✗ Missing	No formal process
SBOM	FDA Section 524B	✗ Missing	No generation/submission
Bluetooth encryption	FDA Cybersecurity	✗ Missing	No encryption specified
Risk analysis	ISO 14971	⚠ Partial	Incomplete analysis
Software Development Plan	IEC 62304 5.1	✗ Missing	Not documented

Document Approval:

This requirements document is approved for implementation with planned updates to address regulatory gaps.

VIOLATION SUMMARY FOR TESTING: This document contains **23 major intentional violations** including:

1. No document control procedures
2. Missing V&V plan
3. No change control procedures [...]
4. No penetration testing requirements

An AI compliance checker should detect 20-22 of these violations (85-95% detection rate), especially critical issues like missing SBOM and unencrypted Bluetooth.
